

Fall 2024 CS543 / ECE549

Computer Vision



Course webpage URL: <http://slazebni.cs.illinois.edu/fall24>

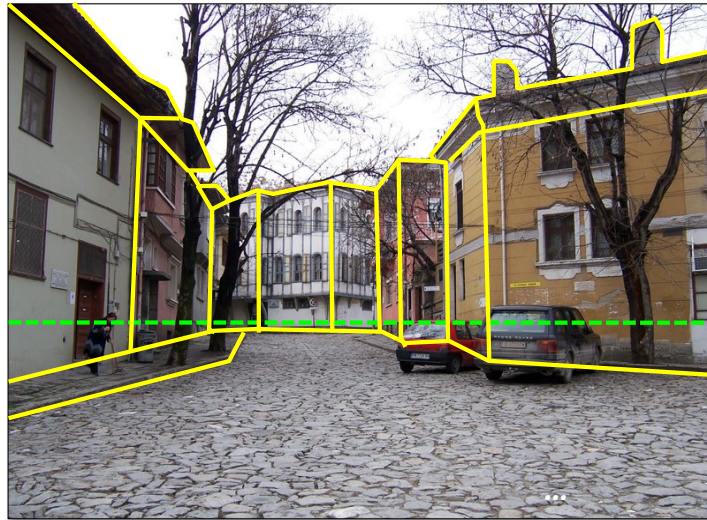
Outline

- Logistics, requirements
- Goal of computer vision
- Origins of computer vision
- Topics covered in class

Goal: Image understanding



Goal: Image understanding



Geometric understanding

Goal: Image understanding



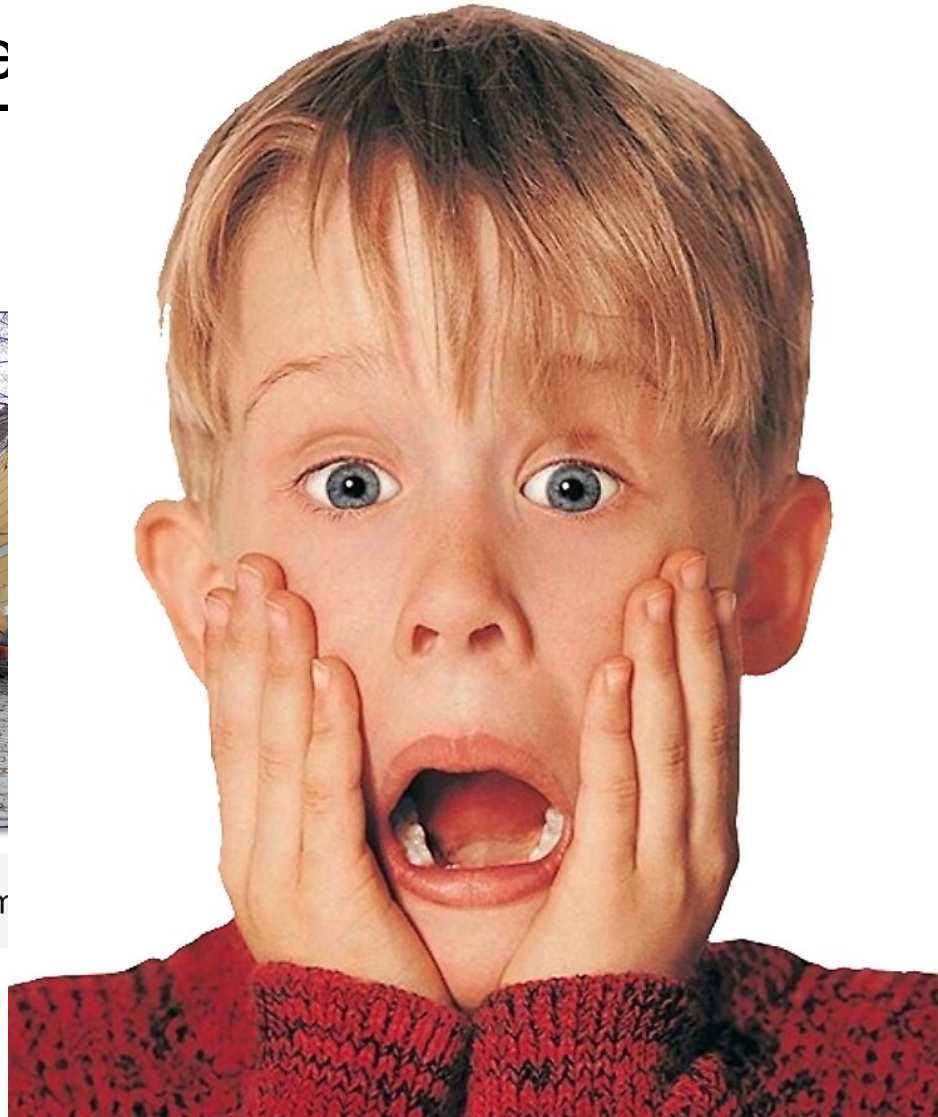
Semantic understanding

Goal: Image

ChatGPT 4o ▾



Please describe this image



It appears to be an older European town or
with a slightly weathered appearance,
streets.

Along the street near a light green building. The
buildings are painted in muted colors, such as light
green, with some cracks and peeling paint.

A different style and color scheme. One
building shows signs of wear and tear on its façade. The
overall look. There are two parked cars in front of

A more ornate building painted white with
a historical and architectural significance,
the street gently curves around this building.

The photo was taken during late autumn or
winter, capturing a moment in time in an older

Goal: Image understanding



Goal: Image understanding



Affordances

Goal: Image understanding



Actionable information

Goal: Image understanding



This image depicts a quiet, cobblestone street in what appears to be an older European town or city. The street is lined with historic buildings that have a slightly weathered appearance, suggesting that they have been standing for many years.

On the left side of the image, there is a person sweeping the street near a light green building. The buildings on this side have old wooden windows and are painted in muted colors, such as light green and a faded pink. The walls show signs of age, with some cracks and peeling paint.

The right side of the street features buildings with a different style and color scheme. One building, in particular, is painted a pale yellow, with visible signs of wear and tear on its façade. The corner of this building is curved, giving it a distinctive look. There are two parked cars in front of this building, one red and the other gray.

In the background, directly ahead, there is a larger, more ornate building painted white with several rows of tall, narrow windows. This building has a historical and architectural significance, with its unique design and intricate window details. The street gently curves around this building.

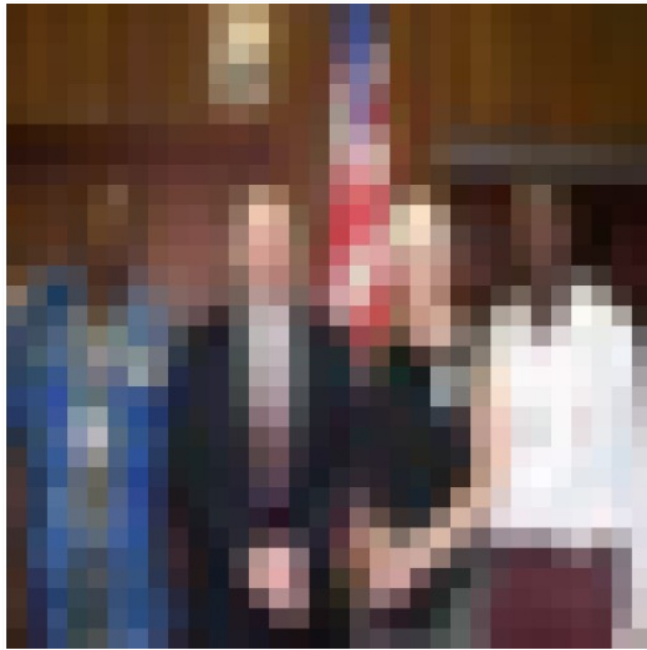
The trees lining the street are bare, indicating that the photo was taken during late autumn or winter. The overall mood of the image is quiet and serene, capturing a moment in time in an older part of town.

Now generate an image using the above text as a prompt



Image generation

Humans are quite good at image understanding



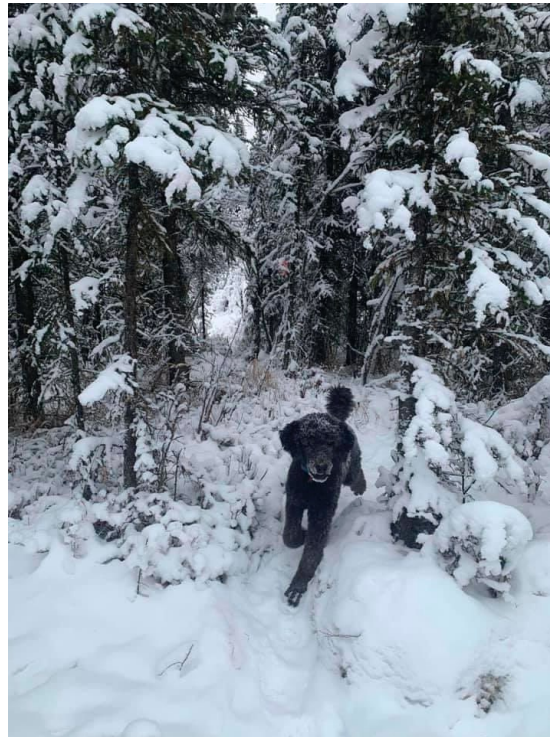
Source: "80 million tiny images" by Torralba et al.

Humans are quite good at image understanding

(most of the time)

Humans are quite good at image understanding

(most of the time)



[Image source](#)

Humans are quite good at image understanding

(most of the time)



Figure from Marr (1982), attributed to R. C. James

Humans are quite good at image understanding

(most of the time)



[What color is this dress?](#)

Remember: Images are fundamentally ambiguous!



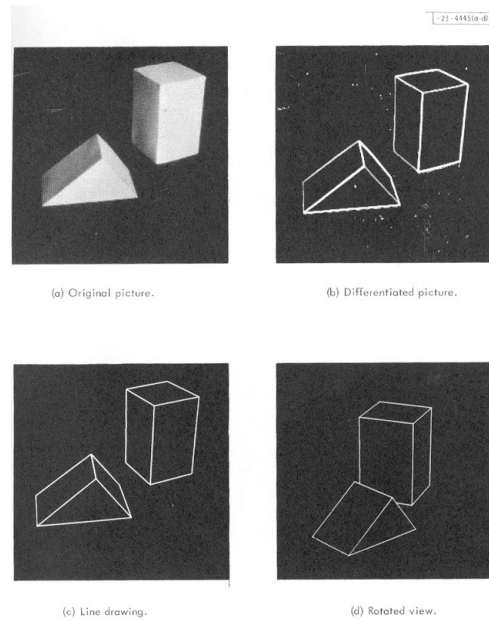
Outline

- Logistics, requirements
- Goal of computer vision
- **Origins of computer vision**

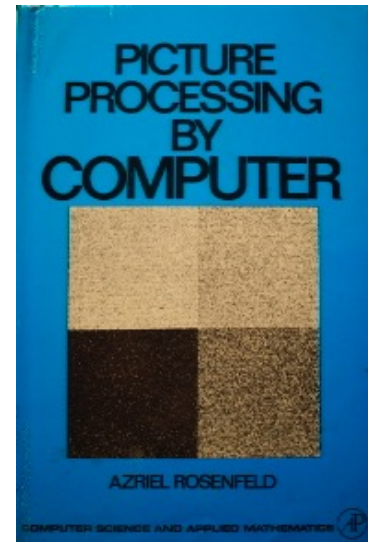
Origins of computer vision



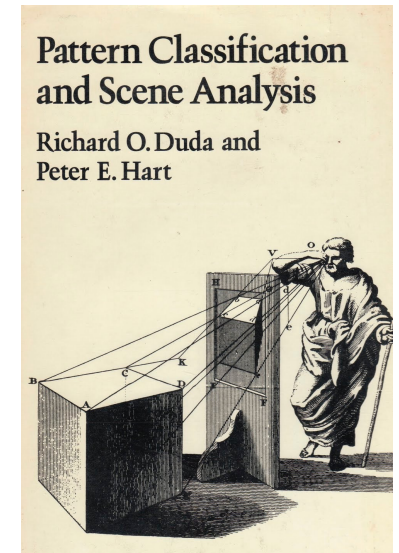
[Hough, 1959](#)



[Roberts, 1963](#)



Rosenfeld, 1969



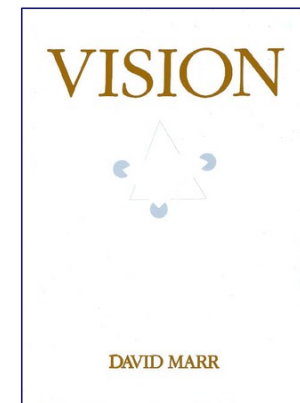
Duda & Hart, 1972

Origins of computer vision

The first great revelation was that the problems are difficult. Of course, these days this fact is a commonplace. But in the 1960s almost no one realized that machine vision was difficult. The field had to go through the same experience as the machine translation field did in its fiascoes of the 1950s before it was at last realized that here were some problems that had to be taken seriously. The reason for this misperception is that we humans are ourselves so good at vision. The notion of a feature detector was well established by Barlow and by Hubel and Wiesel, and the idea that extracting edges and lines from images might be at all difficult simply did not occur to those who had not tried to do it. It turned out to be an elusive problem: Edges that are of critical importance from a three-dimensional point of view often cannot be found at all by looking at the intensity changes in an image. Any kind of textured image gives a multitude of noisy edge segments; variations in reflectance and illumination cause no end of trouble; and even if an edge has a clear existence at one point, it is as likely as not to fade out quite soon, appearing only in patches along its length in the image. The common and almost despairing feeling of the early investigators like B.K.P. Horn and T.O. Binford was that practically anything could happen in an image and furthermore that practically everything did.

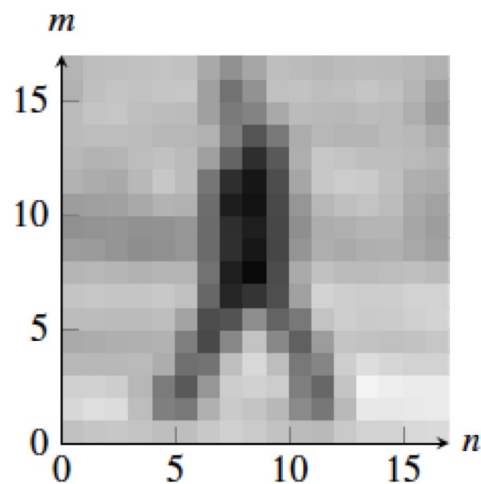


[David Marr \(1945-1980\)](#)



Goal of this class: To get to know the pixels

What we see



What the machine gets

$I =$

160	175	171	168	168	172	164	158	167	173	167	163	162	164	160	159	163	162
149	164	172	175	178	179	176	118	97	168	175	171	169	175	176	177	165	152
161	166	182	171	170	177	175	116	109	169	177	173	168	175	175	159	153	123
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177	178	176	173	174	180	150	27	101	94	74	189	188	186	183	186	188	187
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184	187	192	182	124	32	109	168	171	167	163	51	105	203	209	203	210	205
191	198	203	197	175	149	169	189	190	173	160	145	156	202	199	201	205	202
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Source: [A. Efros](#)

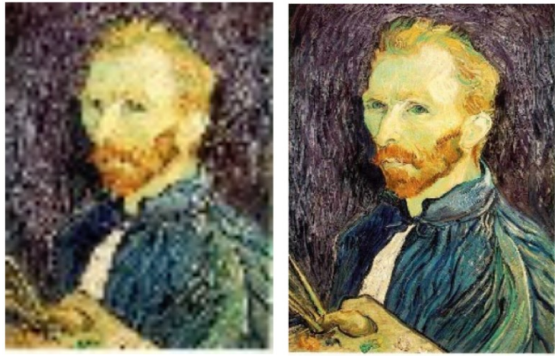
Outline

- Logistics, requirements
- Goal of computer vision and why it is hard
- Origins of computer vision
- Topics covered in class

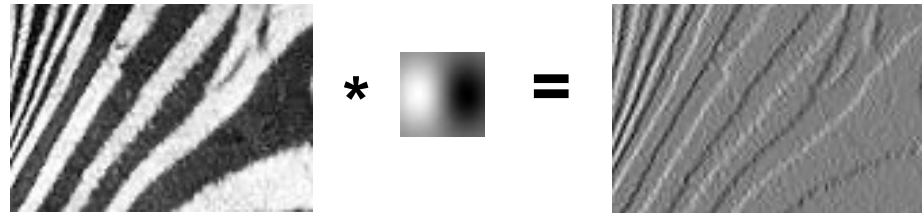
Topics covered in class

- I. Image processing and early vision
- II. Fitting and alignment
- III. Image formation
- IV. 3D vision

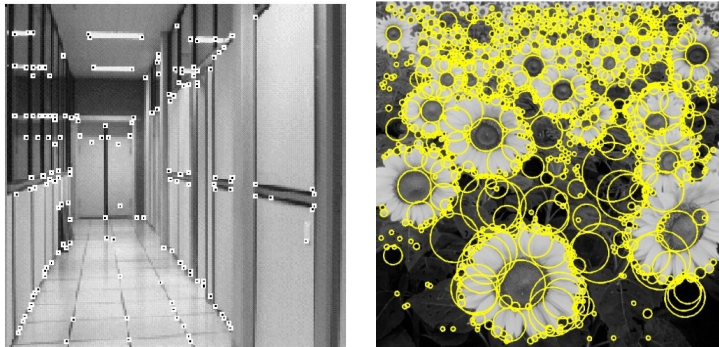
I. Image processing and early vision



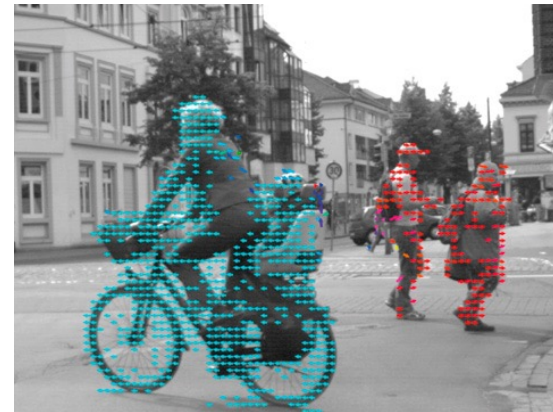
Sampling, interpolation, Fourier analysis



Linear filtering
Edge detection

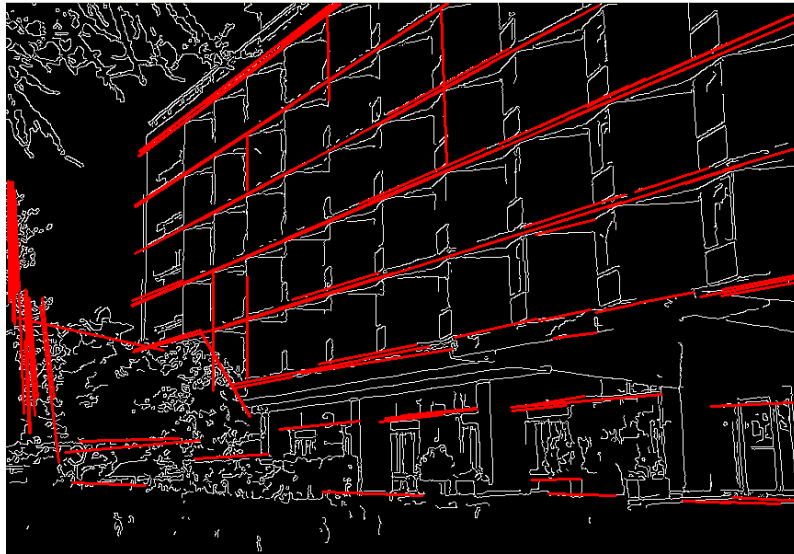


Feature extraction



Optical flow

II. Fitting and alignment

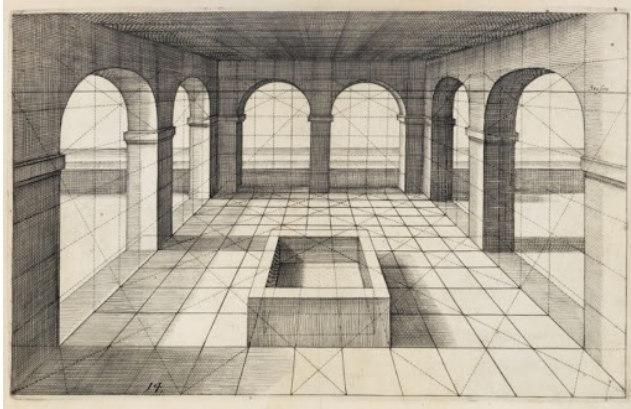


Fitting: Least squares, voting methods



Alignment, image stitching

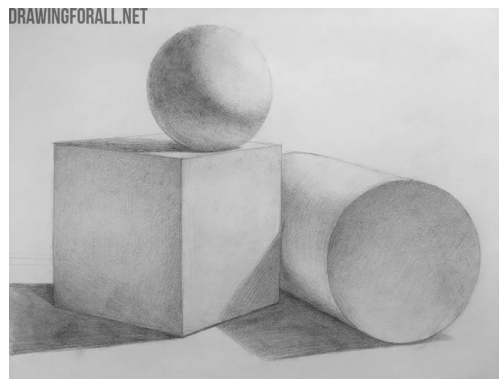
III. Image formation



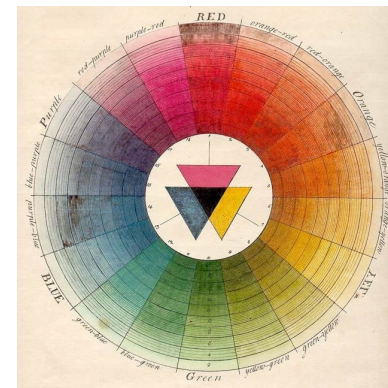
Perspective projection



Camera optics



Light and shading

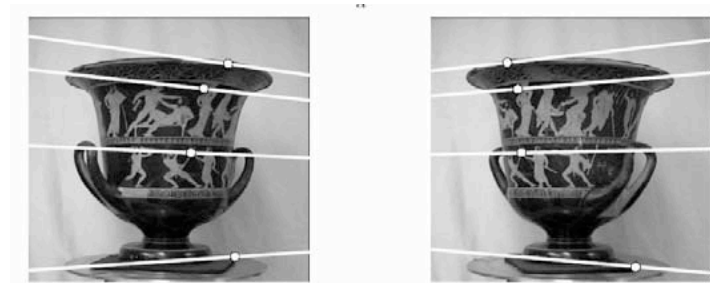


Color

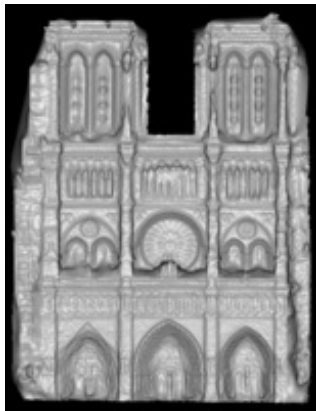
IV. 3D vision



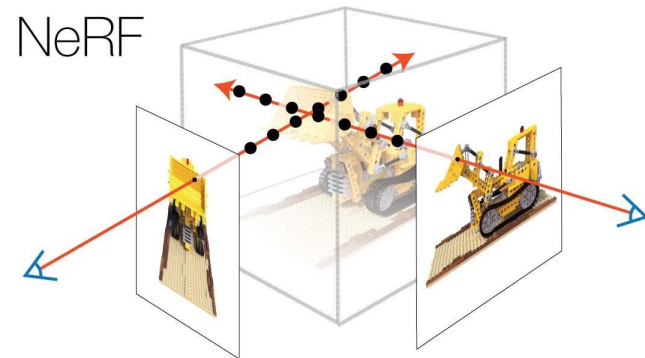
Camera calibration



Two-view geometry, structure from motion

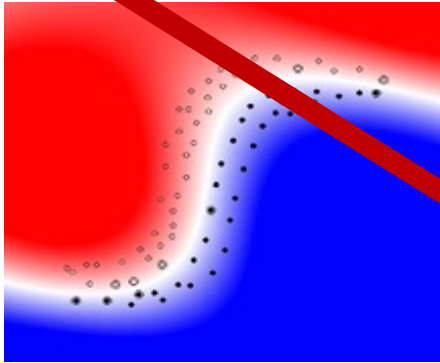


Stereo



Light fields and dense 3D reconstruction

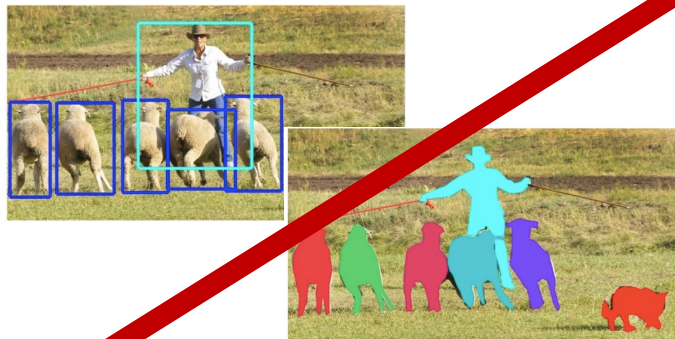
Deep learning stuff



Intro to classification
and neural networks



Deep learning architectures for images



Object detection and segmentation

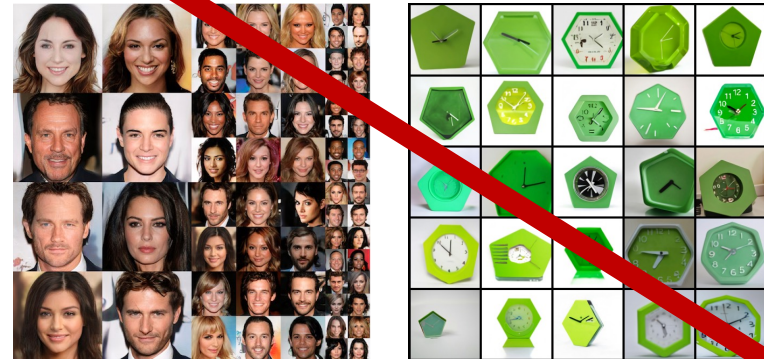


Image generation